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# Supplementary feeding as a source of multiresistant *Salmonella* in endangered Egyptian vultures

**Guillermo Blanco**

Department of Evolutionary Ecology, Museo Nacional de Ciencias Naturales (CSIC), Madrid, Spain

**Correspondence**
Guillermo Blanco, Department of Evolutionary Ecology, Museo Nacional de Ciencias Naturales (CSIC), Madrid, Spain.
Email: gblanco@csic.es

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**Summary**
Wild birds have repeatedly been highlighted as vectors in the dissemination of livestock and human pathogens. Here, the occurrence, serotypes and antimicrobial resistance of *Salmonella* were assessed in adult Egyptian vultures (*Neophron percnopterus*), to test the hypothesis that infection is associated with the consumption of swine carcasses provided at supplementary feeding stations (SFSs). Faeces of year-round resident griffon vultures (*Gyps fulvus*) were also tested to assess whether infection was acquired in the breeding grounds of both species or in the African wintering quarters of Egyptian vultures. Depending on the shedding rate criteria considered, the occurrence of infection in Egyptian vultures varied between the three consecutive sampling days in a range with a minimum of 23%–41% and a maximum of 64%–92% of individuals ($n$ = 11–14 individuals, 27–39 faeces). The occurrence in the single sampling of griffon vultures was 61% of faeces ($n$ = 18). Vultures mostly fed on pig carcasses, which together with their predominant infection with multiresistant serotypes (mostly the monophasic 4,12:i:- variant resistant to aminopenicillins, aminoglycosides and tetracyclines) typically found in pigs from Spain, strongly supports a carcass-to-vulture transmission and cross-infection routes at SFSs. Efforts are encouraged to avoid discarding carcasses of pigs with *Salmonella* at SFSs established for the conservation of threatened scavengers. This could contribute to reducing the long-distance transmission of resistant pathogens with an impact on livestock and human health while avoiding infection risk and its effects on wildlife.

KEYWORDS
feeding stations, multiresistant serotypes, pig carcasses, *Salmonella*, vultures

## 1 | INTRODUCTION

Food and foraging conditions of many wildlife species are being profoundly modified by human activities around the world (Fischer & Lindenmayer, 2007; Sih, Ferrari, & Harris, 2011). Unintended and deliberate feeding of scavenger raptors with anthropogenic foodstuffs includes a variety of remains from domestic refuse and livestock concentrated in rubbish dumps and supplementary feeding stations (SFSs) for the conservation of these species (Cortés-Avizanda et al., 2016). These superabundant and predictable food sources often promote changes in the diet, behaviour, distribution, density and health of wildlife that can exert direct influences on their demography and population dynamics (Cortés-Avizanda et al., 2016; Kubasiewicz, Bunnefeld, Tulloch, Quine, & Park, 2016; Robb, McDonald, Chamberlain, & Bearhop, 2008).

The accumulation of decomposing refuse and carrion is often associated with many infectious organisms infecting domestic animals and humans (Friend & Franson, 1999; Thomas, Hunter, & Atkinson, 2007). Wild birds have repeatedly been cited as vectors in the dissemination of these pathogens (e.g., Fallacara, Monahan, Morishita, & Wack, 2001; Tsiodras, Kelesidis, Kelesidis, Bauchinger, & Falagas, 2008). However, few studies have specifically highlighted the fact that the elimination of residues from human activities is generally at the root of the problem of pathogen dispersion attributed to birds (Friend, McLean, & Dein, 2001). Importantly, birds can be infected with pathogens proliferating in refuse and carcass dumps, and with emerging livestock pathogens associated with intensive or unsanitary farming operations from which the remains discarded there originate (Blanco, Junza, & Barrón, 2017a; Friend & Franson, 1999; Pitarch, Gil, & Blanco, 2017; Tsiodras et al., 2008). These types of food sources can also play an indirect role in wildlife health through physiological changes and contamination making them more susceptible to the acquisition, transmission and effects of pathogens (Arnsberg, Skorping, Grenfell, & Read, 1998; Becker, Streicker, & Altizer, 2015; Blanco et al., 2017a; Lebarbenchon, Poulin, Gauthier-Clerc, & Thomas, 2006; Pitarch et al., 2017). These pathogens can show resistance to the antibiotics and other drugs used to combat them, which is one of the main public health concerns worldwide (Levy, 2002). Bacteria from free-living birds can acquire resistance to antibiotics due to multiple contaminations with human and livestock residues, and disperse this resistance around the world (Allen et al., 2010). Specifically, by consuming carcasses of treated livestock, scavengers can be directly exposed to antibiotics and other drugs used to combat disease (Blanco, Junza, & Barrón, 2017b; Blanco, Junza, Segarra, Barbosa, & Barrón, 2016; Blanco et al., 2017a), and then acquire or develop pathogen resistance to these drugs (Blanco, 2015). Therefore, the combined evaluation of diet, occurrence of pathogens and their drug resistance can provide information about the scavengers' exposure to food acting as infection sources. This can provide insight into public health and wildlife conservation with multiple implications.

*Salmonella* species are food-borne zoonotic pathogens of major public health concern, and leading to huge economic costs from animal losses and human illnesses throughout the world (Hilbert, Smulders, Chopra-Dewasthaly, & Paulsen, 2012; Majowicz et al., 2010; Thorns, 2000; Wray & Wray, 2000). Wildlife can act as long-term asymptomatic carriers of *Salmonella*, but this bacterium can also lead to mortality in weak individuals and cause outbreaks affecting large proportions of populations (Alley et al., 2002; Hall & Saito, 2008; Phalen et al., 2010; Refsum, Vikoren, Handeland, Kapperud, & Holstad, 2003). Negative effects in affected survivors include prominent and persistent gastroenteritis associated with intestinal inflammation and diarrhoea, infertility, as well as a negative impact on breeding success by effecting embryonic and neonatal mortality (Battisti, Giovanni, Agrimi, & Bozzano, 1998; Benskin, Wilson, Jones, & Hartley, 2009; Tizard, 2004). Necrophagous birds can be directly infected through the ingestion of *Salmonella* present in animal carcasses (Friend & Franson, 1999; Tizard, 2004). Infection may be enhanced in these species because they consume the digestive tract and the undigested and faecal content of carcasses as well as excreted faeces of live animals (Blanco et al., 2013; Negro et al., 2002), which is the normal habitat of this bacterium with faecal–oral transmission strategies (Monack, 2011). Knowledge on the prevalence and serotypes of *Salmonella* present in wild populations of obligate and facultative necrophagous birds feeding on livestock carcasses in SFSs is reduced to the medium-high prevalence of the monophasic serotypes of *Salmonella Typhimurium* reported in red kites (*Milvus milvus*) and griffon vultures (*Gyps fulvus*) in Spain (Blanco, 2015; Marin, Palomeque, Marco-Jiménez, & Vega, 2014). In these cases, the serotypes of the isolated strains and their antibiotic resistance profile point towards a role of livestock carcasses, especially swine, disposed at SFSs as the source of infection (Blanco, 2015; Marin et al., 2014).

In this study, the occurrence of *Salmonella* was assessed in faeces of adult Egyptian vultures (*Neophron percnopterus*) foraging at a carcass dump used as a SFSs for the conservation of avian scavengers. The objective of this study was to test the hypothesis that faecal shedding of *Salmonella* is associated with the consumption of livestock carcasses provided in the dump. In the study area, avian scavengers obtain much of their food from pig carcasses from factory farms (Blanco, 2014), which are hence expected to be the primary source of *Salmonella* (Andrés et al., 2013; Blanco, 2015). To test this prediction, the diet of Egyptian vultures was evaluated through prey remains and pellets collected in a roosting site used by the individuals sampled for faeces in the same period, and by determining the available carcasses and their consumption in the dump. The origin of the *Salmonella* infections was further assessed by evaluating whether the serotypes present in the vultures can be linked to the livestock species whose carcasses are found in the dump. Specifically, if pig carcasses are the main food and source of *Salmonella*, the strains isolated from scavenger faeces should correspond to serotypes with antibiotic resistance patterns well established as common in this livestock. As the Egyptian vulture is a migratory species wintering in Africa, faeces of griffon vultures present year-round in the area and feeding at the same dump were also evaluated. This between-species comparison aimed to assess whether *Salmonella* infections in Egyptian vultures occurred in the wintering rather than in the breeding quarters. Given that the Egyptian vulture is globally endangered, the results are discussed for their implications in the management of SFSs and their potential negative impact on health and conservation of their target populations.

## 2 | MATERIALS AND METHODS

### 2.1 | Study species and study area

The Egyptian vulture is a small (~1.8 kg), cliff-nesting vulture living in open, arid and rugged landscapes in southern Europe, Africa and Asia. Mainland populations in Europe are trans-Saharan migratory, with wintering areas in the Sahel and tropical Africa. It is listed as globally endangered due to severe declines experienced throughout its range (BirdLife International, 2016). The Spanish population represents the main stronghold in Europe, despite the fact that it has suffered a strong decline due to illegal persecution through poisoning, habitat alteration and reduction of food supplies (Donázar, 2004). The primary food sources for this species are small and medium-sized wild animals and carcasses of livestock (Donázar, 1993).

This study was carried out in central Spain. This area holds numerically important and increasing populations of Griffon vultures that are highly dependent on livestock carrion at SFSs (Blanco, 2014). Contrary to this large vulture species, the Egyptian vulture has suffered a strong decline in the last few decades in the study area (Segovia province), passing from 43 pairs in 1993 to about 25 pairs in the last years (Del Moral & Martí, 2002; Sanz-Aguilar et al., 2017; author unpubl. data). Adult individuals return to the study areas from their African wintering quarters in early March. Established breeders immediately occupy and defend their territories, and temporally use communal roosts shared with non-breeding adults and individuals performing pre-nuptial migration to northern latitudes; communal roosts are generally associated with abundant and predictable source of food, especially SFSs (Donázar, Ceballos, & Tella, 1996).

## 2.2 | Counts of avian scavengers and carcass availability

The number of Egyptian vultures gathering to spend the night in a communal roost used by this species at least during the last 3 years was recorded for five consecutive afternoons in March 2016. On the counting dates, Egyptian vultures had already been in the area for several weeks, coinciding with first arrivals to the study area. The roosting site was located in a small group of poplars (*Populus* sp.) near (~500 m) a SFS where livestock carcasses, mainly of swine from nearby intensive farms, are regularly deposited for scavenger consumption. The movements of Egyptian vultures were visually tracked from foraging areas to pre-roost gathering sites, where they aggregate regularly before arriving and settling in the trees used to roost. The observations and counts were conducted from a vantage point located at a large distance (1.5 km) from the roost using a telescope to avoid any disturbance to the roosting individuals. The counts lasted from about 2 hr before sunset until all the vultures were settled in the trees.

The number of Egyptian vultures and other obligated and facultative avian scavengers exploiting carrion in the carcass dump located close to the roost were counted during visits conducted in the morning (between 10 a.m. and 12 p.m.). During each visit, the presence and number of fresh carcasses of each livestock species were recorded to estimate the occurrence of available carrion, as a proxy of food predictability, and the availability of carrion from each livestock species (Blanco, 2014).

## 2.3 | Collection of faecal samples

For three consecutive days (25, 26 and 27 of March), all fresh faeces (urine plus faecal material) found beneath the trees used by communally roosting Egyptian vultures were collected using sterile gloves. The sampling was conducted for three consecutive days to improve *Salmonella* detection due to the limited sensitivity of microbiological culture-dependent methods on faecal samples, and because infected individuals intermittently shed this bacterium in the faeces (Tizard, 2004).

On the day before the first faeces collection, the area below the communal roosting trees was cleaned by removing old faeces and the surrounding substrate to avoid collecting old faeces, although these faeces were rarely present on the ground beneath the roosts more than 1 day after excretion, due to their rapid desiccation and degradation. All faeces found below the roosting trees were collected after sunrise, waiting until all the vultures left the roost to do not disturb them, on the three sampling days. The roost was exclusively used by Egyptian vultures, and thereby the collected faeces were not confused with those from other species.

Faecal samples of Egyptian vultures were collected on the first ($n = 39$), second ($n = 27$) and third ($n = 30$) consecutive sampling days. The number of faeces collected on each sampling morning was divided by the number of Egyptian vultures in the communal roosts the respective previous night to estimate the number of faeces approximately corresponding to individual vultures. This estimation assumes that individual vultures excrete a similar number of faeces at night.

In addition, a sample of griffon vulture faeces ($n = 18$) was collected. The sampling was conducted in the morning of March 27 (the third day of collection of Egyptian vulture faeces, see above), just after a large number of individuals (>700) left the SFSs after consuming several pig carcass. The faeces were collected in a pasture close (distant 30 m) to the SFSs and used as a resting area by the griffon vultures after consuming the carcasses. The collected faeces were selected based on their location separated from each other by at least 5 m, to avoid multiple samples from the same individual. Given the large number of griffon vultures present in the resting flock, the collected faeces were not confused with those of other scavengers, which are generally displaced by griffon vultures monopolizing the carcasses in large groups (Cortés-Avizanda, Carrete, & Donázar, 2010; Cortés-Avizanda et al., 2016).

Samples were kept refrigerated and transported to the laboratory (Laboratorio Regional de Sanidad Animal, Consejería de Medio Ambiente y Ordenación del Territorio, Comunidad de Madrid) 1–3 days after collection. Once in the laboratory, the samples were processed within one to two hours following their arrival.

## 2.4 | Egyptian vulture diet

All food remains and pellets found beneath the roosting trees were collected and analysed to assess diet. Two collection periods were considered. First, I collected all food remains and pellets corresponding to the period before the faecal sampling for the determination of *Salmonella* (hereafter initial diet sampling). This period ranged from the beginning of the use of the roosting site by Egyptian vultures, just after arriving from the African winter quarters about 20 days earlier (early March), to the first faecal sampling. Second, all the new food remains and pellets corresponding to the diet of Egyptian vultures during the period of faecal sampling were collected (hereafter final diet sampling). This collection was conducted in the morning corresponding to the last (third) faecal sampling (March 27). In this way, diets corresponding to the period just before the beginning of faecal sampling (initial food sampling) and those corresponding to the period of faecal sampling (final diet sampling) were compared to assess the influence of diet on the occurrence of *Salmonella* in faeces.

Collected food remains correspond to food items transported by the vultures to consume in the roosting trees, generally late in the evening just before roosting (pers. obs.). The remains were identified macroscopically and quantified assuming that each item was singly transported to the roosts on each occasion by individual vultures. Given that most remains belong to livestock carcasses (see Results), the smallest possible number of individual carcasses cannot be determined by considering the number, size and anatomical position of the remains of each species. Using food remains, diet was likely biased in favour of the most durable food and did not reflect the importance of smaller animals and pieces of skinned carcasses and viscera of livestock and large game species (Blanco, 1997). Overall, 47 food remains were collected under the roosting trees, 41 of which were included in the initial diet sampling, and six in the final diet sampling (see above).

In addition to food remains, pellets were analysed. Food remains in pellets were identified macroscopically by comparison to reference collections and identification keys. The typical structure of pellets, composed of hair and feathers of consumed animals, makes it difficult to quantify the number of individual items and their variable age-related biomass, especially from livestock carcasses. Therefore, as in other studies analysing scavenger pellets, the percentage of pellets in which each kind of food occurred was considered to quantify the diet (Blanco, 2014; Donázar, Cortés-Avizanda, & Carrete, 2010). Overall, 115 pellets were collected under the roosting trees, 95 in the initial diet sampling, and 20 in the final diet sampling.

To assess broad differences in diet between initial and final sampling, animal species identified from food remains or pellets were grouped into three categories corresponding to pigs, other livestock species and wild animals.

## 2.5 | *Salmonella* isolation and characterization

All faecal samples were analysed at Laboratorio Regional de Sanidad Animal, Consejería de Medio Ambiente, Administración Local y Ordenación del Territorio, Comunidad de Madrid, Colmenar Viejo, Madrid, Spain), which is accredited by ENAC (National Accreditation Entity) according to the international laboratory standard ISO 17025 for testing and calibration laboratories.

Samples were cultured according to ISO 6579:2002/Amd. 1:2007. Briefly, pre-enrichment was carried out in buffered peptone water (BPW) (Bio Merieux, Marcy l'Etoile, France) for 16 hr at 37°C, and then 100 μl was cultured in Rappaport-Vassiliadis soya peptone (RVS) and 100 μl in modified semisolid Rappaport-Vassiliadis (MSRV) media (Bio Merieux, Marcy l'Etoile, France) for 24 hr at 41.5°C and subcultured onto solid media SM ID agar and XLD agar for 24 hr at 37°C. Suspect colonies were then subcultured onto nutrient agar, and confirmation of *Salmonella* species was performed using oxidase, API 20E and polyvalent antisera.

Serotyping was carried out according to the Kauffmann-White-Le Minor scheme at the Spanish National *Salmonella* Reference Laboratory (Laboratorio Central de Veterinaria, MAGRAMA, Algete, Madrid, Spain).

## 2.6 | Antimicrobial agent susceptibility testing

Antibiotic susceptibility of *Salmonella* to selected antimicrobials commonly used in livestock farming and human medicine was tested using the Kirby-Bauer disc diffusion method, which was performed and interpreted as described by the Clinical and Laboratory Standards Institute (CLSI) protocols (CLSI, 2008). Antimicrobial discs were applied to the surface of Mueller Hinton agar (Oxoid, Basingstoke, UK). Commercial antimicrobial discs (BD BBL™ Sensi-Disc™, NJ, USA) were used for a total of 11 tested antimicrobials classified by families for determining the main antibiotic resistance profiles, including second-generation fluoroquinolones (ciprofloxacin), third-generation cephalosporins (ceftazidime), aminopenicillins (ampicillin, amoxicillin, amoxicillin–clavulanic acid), aminoglycosides (streptomycin, gentamicin, neomycin), sulphonamides (sulfamethoxazole-trimethoprim) and tetracyclines (tetracycline, doxycycline). Each *Salmonella* isolate was classified as susceptible, intermediate or resistant, depending on the growth inhibition diameter according to the CLSI guidelines. Concentration of antibiotics on the discs used, and cut-off values representing the maximum inhibition zone diameter for considering an isolate as resistant were shown in Blanco (2015). Quality control of microorganisms and culture medium was carried out according to ISO 11133, and reference strains of *Salmonella* Enteritidis CECT 4300 (ATCC 13076; ATCC 25928) and) and *Escherichia coli* (CECT 516 = ATCC 8739) were used as controls for each experiment (CECT, Universidad de Valencia, Spain) to ensure the acceptability of results.

# 3 | RESULTS

## 3.1 | Use of the SFSs by scavengers during faecal collection

During the study period, 12 carcasses of large pigs were available for scavenger consumption in the SFSs. Eight of the carcasses were already partially consumed at the beginning of the study and were progressively consumed further afterwards, while the remaining four carcasses of entire large pigs were disposed of on the last sampling day. In addition, the bones and skin corresponding to about 150 old carcasses of pigs and 10 old carcasses of sheep were present in the SFSs, as estimated by counting the remaining craniums.

The SFSs were regularly used by a rich community of facultative and obligate scavengers, mostly raptors (Accipitridae) and corvids (Corvidae). The number of individuals of each species on each sampling day was variable and especially high in the case of griffon vultures after the disposal of fresh livestock carcasses (Table 1).

The roosting site was used by 11–19 Egyptian vultures during the sampling period (Table 1). A high proportion of these individuals

**TABLE 1** Number of individuals of each scavenger species observed in the morning (between 10 a.m. and 12 p.m.) of five consecutive days in a SFS mostly supplied with swine carcasses in central Spain

| | Consecutive observation days | | | | |
|---|---|---|---|---|---|
| Scavengers species | A | B | C | D | E |
| Egyptian vulture (*Neophron percnopterus*) | | | | | |
| Counts in the SFSs (morning) | 21 | 5 | 6 | 6 | 6 |
| Counts in the roost | 19 | 14 | 12 | 11 | 15 |
| Other species: counts in the SFSs (morning) | | | | | |
| Griffon vulture (*Gyps fulvus*) | 5 | 25 | 15 | 5 | >700 |
| Cinereous vulture (*Aegypius monachus*) | 2 | 1 | 1 | 3 | 3 |
| Red kite (*Milvus milvus*) | 2 | 7 | 1 | 1 | 5 |
| Black kite (*Milvus migrans*) | 2 | 3 | 1 | - | 1 |
| White stork (*Ciconia ciconia*) | 1 | - | 2 | - | 2 |
| Raven (*Corvus corax*) | - | 3 | 6 | 2 | 15 |
| Carrion crow (*Corvus corone*) | 2 | - | 3 | 5 | - |
| Eurasian magpie (*Pica pica*) | 1 | 1 | 5 | 2 | 2 |

Data for Egyptian vultures included observations in the morning, as well as counts in the nearby communal roost associated with the SFSs. First observation day corresponds to 23 March 2016 (A) while the last corresponds to 27 March 2016 (E).

foraged in the nearby SFSs during the evening, which was ascertained by observing them while feeding on livestock carcasses and following them from the SFSs to the night roost. These observations were enhanced by the short distance between the SFS and the roosting site. The number of Egyptian vultures exploiting carrion in the SFSs in the morning was higher on the first sampling day, and lower than those counted in the roost on each of the following sampling days (Table 1).

## 3.2 | Diet

The analysis of prey remains and pellets showed that Egyptian vultures using the communal roost fed mostly on swine (93.6% of prey remains, $n$ = 47; 100% of pellets, $n$ = 115). Other much less frequent remains found below the roosting trees and in pellets corresponded to sheep and poultry among other livestock (4.3% of remains, $n$ = 47; 0.9% of pellets, $n$ = 115), and to Roe deer (*Capreolus capreolus*), Wild boar (*Sus scrofa*) and Iberian hare (*Lepus granatensis*) among wild animals (2.1% of remains, $n$ = 47; 10.4% of pellets, $n$ = 115, Figure 1). The proportion of swine, other livestock and wild animals, was similar between the initial and final sampling of food remains and pellets (Fisher exact test, both $p$ > .34), due to the predominant role of swine throughout the study period (Figure 1).

## 3.3 | *Salmonella* shedding and serotypes

A proportion ranging from 23% to 41% of faeces collected on each of the three consecutive days were positive to *Salmonella* (Figure 2).

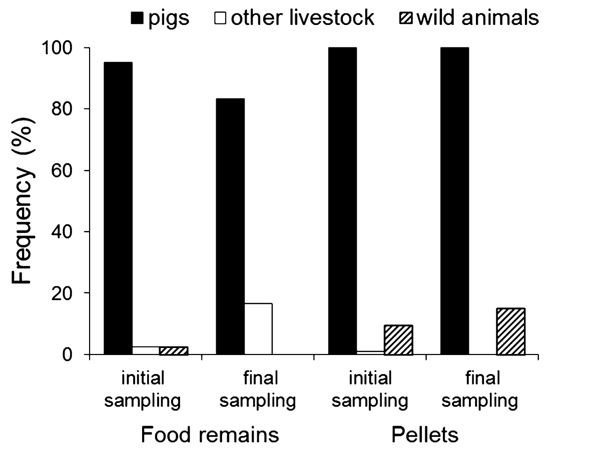


**FIGURE 1** Diet of Egyptian vultures evaluated through prey remains and pellets collected in a roosting site used by individuals foraging in the monitored SFSs during the faecal sampling period (final sampling) and just before (initial sampling). Food from remains and pellets was categorized as corresponding to pigs, other livestock and wild animals

The proportion of *Salmonella*-positive and *Salmonella*-negative samples did not differ between the sampling days (Fisher exact test, $p$ = .33).

The number of faeces excreted ranged between two and three faeces per night and individual vulture (Table 2). The number of *Salmonella*-positive faeces divided by the estimated number of faeces excreted per individual rendered a minimum proportion ranging from 23% to 41% of vultures shedding *Salmonella* (Table 2). The maximum proportion of individuals shedding *Salmonella* ranged from 64% to 92% depending on the sampling day (Table 3), assuming that each *Salmonella*-positive excreta corresponded to a different individual.

Three serotypes of *Salmonella enterica* subspecies *enterica* were recovered at similar proportions between sampling days (Fisher's exact test with Freeman-Halton extension, $p$ = .17) (Table 3). The Typhimurium monophasic 4,12:i:- was the most frequent serotype in the three sampling days (75% of isolates, pooling all positive

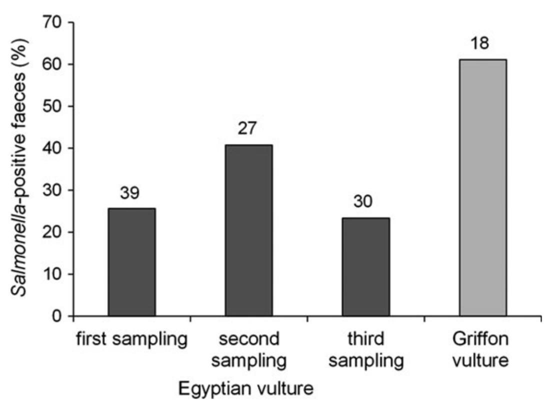


**FIGURE 2** Proportion of *Salmonella*-positive faeces of Egyptian vultures (collected on three consecutive days) and griffon vultures feeding at a SFS in central Spain. Numbers above the bars represent sample sizes (number of analysed faeces)

**TABLE 2** Average number of faeces excreted per individual Egyptian vulture, computed by dividing the vultures attending the roost the night before the sampling of faeces on the morning of three consecutive days

| Faecal sampling (day[a]) | No. Egyptian vultures | No. faeces | Faeces/indiv. | No. faeces positive to *Salmonella* | Estimated number of Egyptian vultures shedding *Salmonella* | |
|---|---|---|---|---|---|---|
| | | | | | Minimum No.[b] (%) | Maximum No.[c] (%) |
| First (B) | 14 | 39 | 2.79 | 10 | 4 (25.6) | 10 (71.4) |
| Second (C) | 12 | 27 | 2.25 | 11 | 5 (40.7) | 11 (91.7) |
| Third (D) | 11 | 30 | 2.73 | 7 | 3 (23.3) | 7 (63.6) |

The number of *Salmonella*-positive faeces divided by the estimated number of faeces excreted per individual rendered a rounded minimum number (and proportion) of vultures shedding *Salmonella*. The maximum number of individuals shedding *Salmonella* was calculated assuming that each *Salmonella*-positive excreta corresponded to a different individual.
[a]Day corresponding to the counts conducted in the roost according to Table 1.
[b]Number of *Salmonella*-positive faeces/number of faeces per individual.
[c]Assuming that each *Salmonella*-positive excreta corresponds to a different individual.

**TABLE 3** Number of strains of each *Salmonella* serotype isolated from faeces of Egyptian vultures feeding on swine carcasses in central Spain

| Faecal sampling (day[a]) Date Serotype | Egyptian vulture First (B) 25/03/2016 *n* (%) | Second (C) 26/03/2016 *n* (%) | Third (D) 27/03/2016 *n* (%) | Total *n* (%) | Griffon vulture 27/03/2016 *n* (%) |
|---|---|---|---|---|---|
| *S. enterica* Typhimurium 4,12:i:1,2 | 4 (40.0) | 1 (9.1) | 1 (14.3) | 6 (21.4) | 3 (27.3) |
| *S. enterica* Typhimurium monophasic 4,12:i:- | 6 (60.0) | 10 (90.9) | 5 (71.4) | 21 (75.0) | 5 (45.5) |
| *S. enterica* Schwarzengrund 4,12:d:1,7 | - | - | 1 (14.3) | 1 (3.6) | - |
| *S. enterica* Israel 9,12:e,h:e,n,z15 | - | - | - | - | 1 (9.1) |
| *S. diarizonae* 61:1,k:1,5,7 | - | - | - | - | 2 (18.2) |

The faeces were collected on three consecutive days in a communal roost close to a SFS. Serotypes of strains isolated from faecal samples of griffon vultures collected in the SFSs are also shown.
[a]Day corresponding to the counts conducted in the roost according to Table 1.

samples, $n = 28$), while the serotype Schwarzengrund 4,12:d:1,7 was only isolated in a sample collected on the last sampling day (Table 3).

About 61% of faeces of griffon vultures collected in the SFSs were positive to *Salmonella* (Figure 2), half of which carried the serotype Typhimurium monophasic 4,12:i:-, while other serotypes were isolated in much lower proportions (Table 3). Serotype Israel 9,12:e, h:e,n,z15 and *diarizonae* 61:1,k:1,5,7 were found in a low proportion of faeces from griffon vultures but not from Egyptian vultures (Table 3).

## 3.4 | Antibiotic resistance

*Salmonella* strains isolated from Egyptian vulture faeces ($n = 23$) were all resistant (100%) to both tetracyclines tested (pooling samples with definitive and intermediate resistance; note that the inhibition test was not conducted in another five *Salmonella*-positive isolates). The isolates were also commonly resistant to ampicillin, amoxicillin, streptomycin (95.7%) and neomycin (82.6%, most of which with intermediate resistance). No *Salmonella* isolates were resistant to the single second-generation fluoroquinolones (ciprofloxacin) and third-generation cephalosporin (ceftazidime) tested; similar results were found in isolates from griffon vulture faeces. Therefore, all strains isolated from both vulture species exhibited resistance to two or more antimicrobial agents (Table 4).

Three and five different patterns of antimicrobial resistance were recorded in *Salmonella* isolates from Egyptian and griffon vultures, respectively (Table 5). The most frequent combination of serotype and resistance profile by antibiotic family was that involving the monophasic 4,12:i:- and multiresistance to aminopenicillins, aminoglycosides and tetracyclines (A-S-T), both in the Egyptian vulture (15 of 23 isolates, 65.3%) and the griffon vulture (three of 10 isolates, 30.0%). Other isolates from this serotype showed resistance to the only sulphonamide tested (sulfamethoxazole-trimethoprim) in addition to aminopenicillins, aminoglycosides and tetracycline (A-S-Su-T). Most isolates of the Typhimurium 4,12:i:1,2 serotype also showed a A-S-T profile of resistance in both scavenger species. In addition, the two *S. diarizonae* 61:1,k:1,5,7 strains recorded in griffon vulture faeces showed a A-S-Su-T resistance profile or only resistance to aminoglycosides (S profile), respectively. Therefore, all *Salmonella* strains were resistant to at least one of the antibiotic families tested, and most were resistant to three or more of these families (Table 5).

# 4 | DISCUSSION

The results from this study support the hypothesis that *Salmonella* infection in Egyptian vultures and other avian scavengers is associated with the consumption of pig carcasses intentionally discarded as supplementary food for these species. A rich community of

**TABLE 4** Antibiotic resistance of *Salmonella* strains isolated from faeces of Egyptian and griffon vultures feeding on swine carcasses at a SFS in central Spain

| | No. (%) of *Salmonella* strains | | | |
|---|---|---|---|---|
| | Egyptian vulture (*n* = 23) | | Griffon vulture (*n* = 10) | |
| Family, agent | Resistant | Intermediate | Resistant | Intermediate |
| Fluoroquinolones (second generation) | | | | |
| Ciprofloxacin | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Cephalosporins (third generation) | | | | |
| Ceftazidime | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Aminopenicillins | | | | |
| Ampicillin | 22 (95.7) | 0 (0) | 8 (80.0) | 0 (0) |
| Amoxicillin | 22 (95.7) | 0 (0) | 8 (80.0) | 0 (0) |
| Amoxicillin–clavulanic acid | 1 (4.3) | 1 (4.3) | 2 (20.0) | 0 (0) |
| Aminoglycosides | | | | |
| Streptomycin | 22 (95.7) | 0 (0) | 8 (80.0) | 1 (10.0) |
| Gentamicin | 4 (17.4) | 6 (26.1) | 2 (20.0) | 0 |
| Neomycin | 3 (13.0) | 16 (69.6) | 1 (10.0) | 6 (60.0) |
| Sulphonamides | | | | |
| Sulfamethoxazole-trimethoprim | 2 (8.7) | 0 (0) | 2 (20.0) | 0 (0) |
| Tetracyclines | | | | |
| Tetracycline | 21 (91.3) | 2 (8.7) | 5 (50.0) | 0 (0) |
| Doxycycline | 21 (91.3) | 2 (8.7) | 6 (60.0) | 2 (20.0) |

The Kirby-Bauer disc diffusion method was performed and interpreted as described by CLSI (2008) protocols.

**TABLE 5** Characterization according to serotype and resistance pattern by antibiotic family of *Salmonella* strains isolated from faeces of Egyptian and griffon vultures feeding on swine carcasses at a SFS in central Spain

| Resistance pattern by antibiotic family[a] | Egyptian vulture | | Griffon vulture | |
|---|---|---|---|---|
| | No. of isolates | Serotype[b] (No. of strains) | No. of isolates | Serotype[b] (No. of strains) |
| A-S-Su-T | 2 | 4,12:i:- (2) | 2 | 4,12:i:- (1), *diarizonae* (1) |
| A-S-T | 20 | 4,12:i:- (15),Typhimurium (5) | 5 | 4,12:i:- (3),Typhimurium (2) |
| S-T | 1 | Schwarzengrund (1) | 1 | Israel (1) |
| A-S | | | 1 | Typhimurium (1) |
| S | | | 1 | *diarizonae* (1) |

[a]A (aminopenicillins), S (aminoglycosides), Su (sulphonamides), T (tetracyclines).
[b]The complete identity of *Salmonella* subspecies and serotypes is shown in Table 3.

facultative and obligate avian scavengers used the monitored SFSs at medium-high abundances, as generally encountered at these locations in the study and other areas (Blanco, 2014; Cortés-Avizanda et al., 2010, 2016). Despite their generally low abundance, and thus their endangered status, a relatively large number of Egyptian vultures foraged daily in the SFSs and roosted nearby. Available carcasses and their consumption in the SFSs, as well as the analysis of pellets and food remains, clearly indicate that Egyptian vultures fed mostly on pig carrion during the faecal sampling days and the period just before, coinciding with their return from wintering quarters in Africa. Among the species recorded in the SFSs, the abundance of griffon vultures was especially high after the disposal of fresh carcasses. The growing dependence of scavengers on swine carcasses from factory farming has been increasingly recorded in several Spanish regions (Camiña & Montelío, 2006; Donázar et al., 2010). This has been shown to be associated with scavenger contamination with veterinary pharmaceuticals used in livestock farming (Blanco et al., 2016; Blanco et al., 2017 a,b) and with infection by parasites and opportunistic and emerging livestock pathogens (Blanco, 2015; Blanco et al., 2017a; Marin et al., 2014; Pitarch et al., 2017).

The presence of culturable *Salmonella* in faeces of Egyptian vultures represents the first report of this bacterium in free-living individuals of this species. The occurrence rate can be considered high in Egyptian vultures and very high in griffon vultures (see also Marin et al., 2014), compared to values reported for apparently healthy free-ranging birds (Benskin et al., 2009; Daoust & Prescott, 2007;

Tizard, 2004). Determining the actual prevalence of *Salmonella* from faeces is challenging due to intermittent shedding by infected individuals, thus generally leading to underestimations when analysing single faeces per individual hosts (Daoust & Prescott, 2007; Tizard, 2004). *Salmonella* detection can also be influenced by the ability of conventional culture methods to detect low levels of the bacterium in faeces (Daoust & Prescott, 2007; Tizard, 2004). To attempt to overcome these difficulties, faeces were collected for three consecutive days, and culture enrichment broths were previously used in the microbiological tests. Although the identity of each individual host excreting each faecal sample was unknown, the similar number of Egyptian vultures attending the roost on the sampling days, and just before and after, suggests that the tested faeces could partially correspond to the same individuals. A partial between-days turnover of individuals using the roost, and the irregular day-to-day and within-day intermittent shedding by particular individuals (Daoust & Prescott, 2007; Tizard, 2004) could explain the slight differences in occurrence among days. Depending on the shedding rate criteria considered, the actual prevalence of infection could vary between days in a range with a minimum of 23%–41% and a maximum of 64%–92% individuals attending the roost. Assuming that *Salmonella*-positive excreta do not always correspond to the same or to different individuals, the actual prevalence would be higher than the minimum occurrence values suggested, but lower than the maximum values reported here. Even in the more conservative scenario, the occurrence of *Salmonella* can be underestimated when using culture methods, as compared to molecular methods detecting partial genomes and dead, unviable or viable but unculturable bacteria from current and past infections (Benskin et al., 2009; Daoust & Prescott, 2007; Tizard, 2004). In any case, the occurrence of culturable *Salmonella* in faeces, as an instantaneous conservative indicator of current infection with potential to be transmitted and cause disease, can be considered high in the studied scavengers.

The results agree with the prediction that *Salmonella* strains isolated from vultures mostly feeding on swine carcasses should correspond to serovars commonly isolated in this livestock. The two Typhimurium serotypes recovered at the highest rates in the two vulture species were among the most frequently reported in fattening pigs in Spain (Arguello, Carvajal, Collazos, García-Feliz, & Rubio, 2012; EFSA-ECDC 2015, 2016; García-Feliz et al., 2007, Vico et al., 2011) and other European countries (EFSA, 2008, 2009; EFSA-ECDC 2015, 2016). In particular, the monophasic variant of *S. typhimurium* was initially characterized from pig samples in Spain, and today is considered an emerging pathogen associated with human and swine infections worldwide (Echeita, Aladueña, Cruchaga, & Usera, 1999; García et al., 2014; Hauser et al., 2010; Hopkins et al., 2010). This serotype was the most commonly shed by Egyptian vultures on the three sampling days, as well as by griffon vultures in a single sampling. This agrees with previous studies also showing that these serotypes were the only or most commonly found in faeces of scavengers feeding on pig carcasses (Blanco, 2015; Marin et al., 2014), and they have also been recorded in other wildlife associated with pig farms in Spain (Andrés et al., 2013; Andrés-Barranco et al., 2014). The similarly common shedding of the two Typhimurium serotypes by Egyptian vultures (present only during the breeding season) and griffon vultures (present year-round) suggests that infections can likely occur in their breeding quarters, where both species continuously share the use of multiple SFSs supplied with swine carcasses in the study area (Blanco, 2014). This is further supported by the low occurrence of serotypes exclusively recovered from Egyptian vultures (Schwarzengrund 4,12:d:1,7) and griffon vultures (Israel 9,12:e,h:e,n,z15 and *diarizonae* 61:1,k:1,5,7). None of these serotypes were among the most commonly reported in livestock populations or food/feed categories in the EU (EFSA-ECDC, 2015). This suggests that strains with exclusive serotypes infrequently recovered in each vulture species could be acquired from infection sources and geographic areas not shared by both vulture species, although more research with larger sample sizes would be required to ascertain this.

Antimicrobial multiresistant strains of zoonotic *Salmonella* can spread from medicated livestock to wildlife, thus creating new reservoirs also acting as amplifiers and long-distance dispersers (Allen et al., 2010). Avian scavengers can be directly affected by the ingestion of antibiotics used to treat livestock (Blanco et al., 2016, 2017a, b), and this can also contribute to the creation of new resistances and the amplification of those acquired from livestock. All *Salmonella* strains isolated from both vultures species showed resistance to two or more of the antimicrobial agents tested. Most strains were simultaneously resistant to aminopenicillins, aminoglycosides and tetracyclines (A-S-T). In particular, the Typhimurium 4,12:i:1,2 and the monophasic 4,12:i:- serotypes with an A-S-T resistance pattern were the most frequent in both species. A lower proportion of the monophasic 4,12:i:- and the *diarizonae* 61:1,k:1,5,7 serotypes showed a A-S-Su-T resistance pattern due to the added resistance to the only sulphonamide tested (sulfamethoxazole-trimethoprim). These specific combinations of serotypes and resistance have been frequently recorded in Spanish swine farms (Mejia et al., 2006; Usera et al., 2002; Vico et al., 2011) and in cases of food-borne human illnesses associated with the consumption of contaminated meat and pork products (Astorga et al., 2007; Echeita et al., 1999; García et al., 2014; Hopkins et al., 2010). Therefore, the antibiotic resistance pattern adds to the evidence provided by the frequency of serotypes, and vulture diet to strongly support the transmission via pig carcasses, as well as the cross-infection and contamination from livestock and scavengers faeces at SFSs. This pig-to-bird transmission route has been previously proposed to explain the concordance of serotypes and resistance patterns of *S. typhimurium* and its monophasic variants at much lower occurrences in a facultative scavenger partially relying on pig carcasses (Blanco, 2015) and passerine birds captured close to pig farms (Andrés et al., 2013; Andrés-Barranco et al., 2014).

In Spain, pig carcasses are almost exclusively available for scavenger wildlife in SFSs (Blanco, 2014). These stations were originally conceived with the intention of contributing to the conservation of vultures (Cortés-Avizanda et al., 2016). Currently, many avian scavengers are increasingly dependent on carcasses of this and other housed livestock due to population reductions of wild animals and free-ranging livestock that constituted their diet in the past (Blanco et al., 2017a; Donázar, 1993; Donázar et al., 2010). An increase in the availability of livestock carcasses from factory farms (especially of swine and poultry) has further favoured the use of concentrated, predictable and abundant food in SFSs, thus promoting the scavengers' habituation to these easily exploited resources (Cortés-Avizanda et al., 2016). This highlights the impact of SFSs in the potentiation of the vulture's role as reservoirs, amplifiers and disseminators of *Salmonella* resistant strains at human-livestock interfaces. Importantly, although generally neglected, infection of enteric food-borne bacteria and other pathogens from livestock can also represent a significant threat to wildlife (Phalen et al., 2010), with implications for the conservation of threatened species (Smith, Sax, & Lafferty, 2006; Thompson, Lymbery, & Smith, 2010).

The potential threat of disease has been suggested to explain the strong decline of the Egyptian vulture (Blanco et al., 2017a; Pitarch et al., 2017), but it is unknown whether *Salmonella* can cause clinical illness in this species. This evaluation would require the assessment of the impact on health, survival and breeding success in long-term monitored vultures with and without *Salmonella* infection. It should be considered that this bacterium can be especially problematic in developing nestlings and immunocompromised individuals, including those affected with other pathogens and contaminants (Daoust & Prescott, 2007). In particular, livestock carcasses may be a shared source of pathogens and pharmaceuticals used to combat them (Blanco et al., 2016, 2017a,b; Pitarch et al., 2017), especially antimicrobials that may alter the normal microbiota and depress the immune system (Keeney, Yurist-Doutsch, Arrieta, & Finlay, 2014). Since *Salmonella* infection may be passed transovarially to eggs, its impact on female infertility and hatching success should also be evaluated in failed eggs to determine its impact on breeding failure and productivity.

The large number of intensive farms of fattening pigs operating in the study area provides a high surplus of carcasses discarded at SFSs (Blanco, 2014). An artificially high predictability and abundance of food provided in fixed locations imply that large numbers of scavengers frequently forage in overcrowding, confinement and unsanitary conditions generally found at SFSs (Blanco, Cardells, & Garijo-Toledo, 2017; Cortés-Avizanda et al., 2016). Therefore, the role of pigs as a primary source of *Salmonella*, and the risk of scavenger infection in SFSs, may continue in the future in the study area. Efforts are encouraged to avoid discarding carcasses of pigs with *Salmonella* at SFSs aimed at the conservation of threatened scavengers. Given the logistic difficulties it can impose, a more practical alternative could be the application of programs promoting low-intensity farming practices and the use of carcasses from free-ranging ruminants left in the countryside for scavengers' consumption. In addition, specific policies could consider forbidding the supply of feeding stations with swine carcasses, as already implemented in other Spanish regions. This could contribute to reducing the long-distance transmission of resistant pathogens with an impact on livestock and human health while avoiding infection risk and its effects on wildlife.


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## CONFLICT OF INTEREST

None conflict declared.

## ORCID

*Guillermo Blanco* http://orcid.org/0000-0001-5742-4929

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